MILITARY SPECIFICATION

PENETRATORS, HULL, CONNECTORIZED, CONNECTORS, PRESSURE-PROOF, FIBER OPTIC, SUBMARINE GENERAL SPECIFICATION FOR

This specification is approved for use by Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fabrication and test of connectorized electrical and fiber optic hull penetrators and pressure-proof connectors intended for use on submarines with electrical and fiber optic cable utilizing one or more single-fiber transmission elements. Compatability with watertight cable as specified in DOD-C-85045 or equivalent and MIL-C-915.

1.2 Classification. Electrical and fiber optic hull penetrator and pressure-proof connectors covered by this specification shall be the following types as specified (see 6.2).

Type I	Pressure-proof connector plugs.
Type II	Pressure-proof connector receptacles.
Type III	Pressure-proof plug caps.
Type IV	Pressure-proof receptacle caps .
Type V	Multi-connector electrical and fiber optic penetrator.
Type VI	Single-connector fiber optic penetrator.
Type VII	Multi-connector fiber optic penetrator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-465	-	Nickel-Aluminum Bronze Alloy.
QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought.
ZZ-R-768	-	Rubber for Mountings (Unbonded-Spool and Compression Types).

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MIL-M-14	-	Molding Plastics and Molded Plastic Parts, Thermosetting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command (SEA 55Z3), Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-I-631	-	Insulation, Electrical, Synthetic Resin Composition, Nonrigid.
MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems Requirements For.
MIL-C-915	-	Cable and Cord, Electrical, for Shipboard Use, General Specifications For.
MIL-G-4343	-	Grease, Pneumatic Systems.
MIL-L-8937	-	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting, NATO Code Number S-1738.
MIL-P-15024	-	Plate, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	-	Plate, Identification.
MIL-P-17091	-	Polyamide (Nylon) Plastic, Rigid: Molded Parts, Rods and Flats.
MIL-W-16878	-	Wire, Electrical, Insulated, General Specification For.
MIL-W-16878/17	-	Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 600 volts, Polymide Jacket.
MIL-E-21562	-	Electrodes and Rods - Welding, Bare, Nickel Alloy.
MIL-E-22200/3	-	Electrodes, Welding, Covered, Nickel Base Alloy and Cobalt Base Alloy.
MIL-B-24059	-	Bronze, Nickel Aluminum; Rod, Flat Products with Finished Edges, Shapes and Forgings.
MIL-C-24231	-	Connectors, Plugs, Receptacles, Adapters, Hull Inserts, and Hull Insert Plugs, Pressure Proof, General Specification For.
MIL-C-24231/11	-	Connector, Multiple Cable.
MIL-C-24231/25	-	Connector, Hull Inserts.
MIL-H-24626	-	Cable Harnesses, Pressure-Proof Fiber Optic, Submarine General Specification For.
MIL-P-25732	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 275°F.
MIL-L-60326	-	Lubricant, Fluorocarbon Telemer Dispersion.
DOD-C-85045	-	Cable, Fiber Optic.
MIL-C-55330	-	Connector, Preparation For Delivery of.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread, Standards for Federal Services.

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MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

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MIL-STD-248	-	Welding and Brazing Procedure and Performance Qualification.
MIL-STD-278	-	Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-810	-	Inspection and Acceptance Standards for Propulsion Fluid Cells and Fittings.
MIL-STD-1344	-	Test Methods for Electrical Connectors.
MIL-STD-45662	-	Calibration Systems Requirements.
MS28775	-	Packing, Preformed, Hydraulic 275°F O-Ring.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B 46.1 - 1962 - Surface Texture. (DOD adopted)

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. All requirements herein apply to all the hull penetrators and connectors, and accessories covered by the applicable specification sheet. Detail requirements or exceptions applicable to the individual types shall be as specified herein or in the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.3, and table I).

3.3 Materials. The electrical and fiber optic hull penetrators, connectors, and accessories shall be constructed of materials as specified herein (see 3.1). All materials used shall be ozone resistant, nontoxic, non-nutrient to fungus, and manufactured to good workmanship quality. Hull penetrator and connector materials shall be selected to provide the degree of immunity to nuclear radiation exposure as specified (see 3.1). Materials which are not specifically described herein shall be approved by the contracting activity. Sufficient material identification and controls shall be exercised by the contractor to assure that the final product conforms to this specification.

3.3.1 Material control objective quality evidence. Objective evidence of chemical analysis and physical properties of material per heat from the mill or per lot from an independent laboratory, traceable from the serial number on the final product, shall be maintained in support of specification compliant material. Chemical and physical properties shall be the result of laboratory analysis. At least one sample from an identifiable heat or individual piece of bar stock is required for analysis. Resulting analysis shall be traceable to and from markings affixed to the material. Material identity, traceable to objective evidence, shall be maintained throughout the production process.

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3.3.1.1 Records of heat treatments. Records of heat treatments shall be maintained. The heat treatments process and serial number of each part which has been heat treated shall be included on these records.

3.3.2 Hull penetrator body, nut and washer. Unless otherwise specified, hull penetrator body, nut and washer, shall be manufactured from nickel-copper-aluminum alloy in accordance with class A of QQ-N-286. After machining the hull penetrator body, nuts and washers shall have the following minimum mechanical properties: 130,000 pound force per square inch (lbf/in^2) minimum tensile strength, 85,000 lbf/in^2 minimum yield strength at 0.2 percent offset, and 20 percent minimum elongation.

3.3.2.1 Age-hardening of nickel-copper-aluminum. When nickel-copper-aluminum bar stock is acquired in the mill annealed condition (annealed by the mill) and manufacturing process is restricted to machining operations, the mill test certification on an age-hardened test coupon may be used to certify mechanical properties of the product provided that identification markings on the finished product establish traceability to the test coupon on which the mechanical tests were determined, and the product is age-hardened by heating to 593°C to 607.2°C and holding 16 hours at temperature, followed by furnace cooling at the rate of 8.3°C to 13.8°C per hour to 482°C or step cooling by furnace cooling to 579°C, holding 6 hours at temperature, furnace cooling to 482°C, holding 6 hours at temperature and then removing from the furnace.

3.3.3 Receptacle bodies. Unless otherwise specified (see 3.1), the receptacle bodies shall be manufactured from nickel-copper alloy conforming to class A, form 1, hot finished in accordance with QQ-N-281, with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy conforming to class A, form 1, cold drawn in accordance with QQ-N-281, with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset. The chemical and physical analysis furnished by the mill shall satisfy certification requirements when supplemented by the required traceability markings identifying the finished receptacle bodies to the bar stock on which the chemical and physical analysis was conducted.

3.3.4 Optical termini. Fiber optic termini shall be as specified (see 3.1).

3.3.5 Molded inserts. Material for molded inserts shall be cast epoxy resin compound or equal.

3.3.6 Plug sleeves. Material for plug sleeves shall be nickel-copper alloy in accordance with class A of QQ-N-281.

3.3.7 Pressure-proof caps and coupling rings. Caps and coupling rings shall be manufactured from nickel-aluminum-bronze in accordance with MIL-B-24059 and QQ-C-465, alloy number C630. Caps are used on multi-pin receptacles (type V).

3.3.8 Protective connector caps. Protective caps shall be plastic type MFI-20 or MAI-60 conforming to MIL-M-14.

3.3.9 O-ring gaskets. O-ring gasket material shall conform to MIL-P-25732.

3.3.10 Retainer plates and retainer plate assemblies. Retainer plates and retainer plate assemblies shall be made from nickel-copper alloy conforming to class A, form 1, hot finished of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy conforming to class A, form 1, cold drawn of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.

3.3.11 Gland nuts. Gland nuts shall be made of nickel-aluminum-bronze conforming to MIL-B-24059 and QQ-C-465, Alloy number C630.

3.3.12 Nylon inserts. Inserts shall be made of nylon conforming to type I of MIL-P-17091.

3.3.13 O-ring lubrication. O-rings shall be lubricated at assembly with a light film of lubricant conforming to MIL-G-4343.

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3.3.14 Designation and information plates. Designation and information plates shall be type B or C (corrosion-resisting steel), or type H (aluminum alloy) of MIL-P-15024, .062 inch (1.57 mm) thick. Plates shall be designed for severe service in accordance with MIL-P-15024/5.

3.3.15 Packing. Packing shall be made of rubber conforming to ZZ-R-768 with a Shore A hardness of 55 ±5.

3.3.16 Fiber optic cable. Fiber optic cables connected to pressure-proof plugs and hull penetrators shall be watertight in accordance with DOD-C-85045.

3.3.17 Screws. Screws used to fasten identification plates shall be fabricated from corrosion resistant steel.

3.3.18 Insert polarizing pin. The pin shall be fabricated from nickel-copper alloy in accordance with class A QQ-N-281.

3.3.19 Electrical pin and socket contacts. Pin contacts shall be Bendix Electrical Components Division, Sidney, NY, size M short, part no. 10-101515-15P, gold-plated or L.L. Rowe Company, Woburn, MA, part no. 40096 or equal. Any other pin contact submitted as equal shall be subject to specific approval of NAVSEA. Socket contacts shall be Bendix Electrical Components Division size 16M short, part no. 10-101515-15S, gold-plated or L.L. Rowe Company part no. 40097 or equal.

3.3.20 Electrical cable. Electrical cables connected to pressure-proof plugs and hull penetrators shall be water-tight in accordance with MIL-C-915 or as approved by NAVSEA.

3.3.21 Socket contact sleeving. Socket contact sleeving shall be type F, subform va, grade A, class II, category 1, and color clear, in accordance with MIL-I-631. Unless otherwise specified (see 3.1) the size of the sleeving shall be 0.118 inch (.298 cm) inside diameter and .562 inch (1.43 cm) length.

3.3.22 Penetrator cables and wires.

3.3.22.1 Cables. Cables shall be the type as specified (see 3.1 and 6.2).

3.3.22.2 Wires. Wires shall be 16 awg stranded tin coated copper in accordance with MIL-W-16878 and MIL-W-16878/17, color black.

3.4 Design, construction, and physical dimensions. Unless otherwise specified (see 3.1), the design, construction and physical dimensions of the penetrators, connectors, and accessory assembles shall be as specified herein.

a. Tolerances shall be as follows:

Under 6 in. (15.24)			6 in. - 24 in. (152.4 - 604.6)		
1 Place	2 Place	3 Place	1 Place	2 Place	3 Place
±.1 (±2.54)	±.02 (±0.51)	±.005 (±0.13)	±.1 (±2.54)	±.04 (±1.02)	±.010 (±0.26)

NOTES:
1. Decimals dimensions.
2. Angles ±0.5 degrees.
3. Chamfer angles ±5 degrees or as specified on drawings.
4. Millimeters are in parentheses.

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b. All sharp corners and edges shall be rounded or chamfered .005 inch to 0.015 inch (0.13 to 0.39 millimeters) maximum, unless otherwise specified (see 3.1).

c. Drilled and tapped holes shall have all burrs removed.

d. Machined surfaces shall be 63 rhr maximum, unless otherwise specified (see 3.1).

e. All machining of metal parts shall be accomplished before plating or other finish is applied.

f. Unless otherwise specified, (see 3.1 and 6.2) threads shall be American National Standard RH class 2. Threads shall be checked by means of ring and plug gages only in accordance with FED-STD-H-28.

g. All surfaces in contact with automatic squeeze type gaskets (such as O-ring) shall have a 32 Rockwell hardness C-5560 maximum finish. These surfaces shall be free of tool marks, nicks and scratches.

h. Prior to welding or brazing, all parts shall be jigged or clamped to insure proper alinement.

i. All cast or molded parts shall be free from cold shuts, blow holes, or other imperfections. All surfaces shall have fins, flashing, and burrs removed.

j. All welding of K-Monel shall be accomplished in the age-hardened condition.

k. All threads of nickel-copper-aluminum parts shall be coated with a dry lubricant per MIL-L-60326, type I prior to assembly; or MIL-L-8937, form B.

l. All surfaces shall be in accordance with ANSI B 46.1.

m. All welders shall be qualified in accordance with MIL-STD-248 and all welding shall be performed in accordance with MIL-STD-278.

n. In welding nickel-copper alloy to nickel-copper-aluminum alloy, the following electrodes shall be used:

(1) 8N12 or 9N10 covered electrode in accordance with MIL-E-22200/3. (Manual metal arc)

(2) MIL-EN 60 or MIL-RN 60 electrode in accordance with MIL-E-21562 (Tungsten Inert Gas).

(3) In accordance with requirement 5 of MIL-C-24231/11.

o. There shall be no weld repair to any portion of a hull penetrator body, cover, nut or washer.

3.4.1 Fiber optic termini. Termini shall conform to details as specified (see 3.1), and shall not be damaged during mating of counterpart connector plugs and receptacles.

3.4.2 Penetrator assembly. The penetrator assembly shall consist of a penetrator body and cover (as applicable), receptacle, receptacle insert assembly (including fiber optic and electrical cables), (as applicable), identification plate, screws, washer, nut, O-ring gaskets, packing, glad nut, retainer plates, protective caps, and retainer rings.

3.4.3 Plug assembly. The plug assembly shall consist of a plug sleeve, polarizing key or keyway, coupling ring, termini contact(s), O-ring gasket, and protective cap.

3.4.4 Receptacle assembly. The receptacle assembly shall consist of a receptacle body, polarizing key or keyway, nylon insert for locking the coupling ring, O-ring gaskets, termini contacts(s), molded receptacle insert with fiber optic or electrical cable, (as applicable), protective cap, and retaining rings, as applicable.

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these units to be subjected to GROUP II tests transmitters with cable pigtails

3.4.5 Pressure-proof receptacle cap. The cap shall consist of a cover which provides for the pressure-proof integrity of the receptacle internals.

3.4.6 Pressure-proof plug cap. The cap shall consist of a cover and O-ring gasket, as applicable, which provides for the pressure-proof integrity of the plug internals.

3.4.7 Markings. The manufacturer's marking, consisting of the military part number, shall appear on the outer diameter of each multiple penetrator body, single penetrator body, and receptacle body. The marking shall be permanent, legible, and shall be applied so as not to induce detrimental stresses or to cause malfunction of the part. Lettering shall be at least (.125 inch (3.18 mm)) high.

3.4.7.1 Marking of nut and washer. Manufacturer's marking, consisting of the military part number, shall appear on the exterior section of each nickel-copper-aluminum alloy nut and washer. The marking shall be permanent, legible, and shall be applied so as not to induce detrimental stresses or to cause malfunction of the part. Minimum height lettering (.125 inch (3.18 mm)) shall be used.

3.4.7.2 Marking of protective caps. The protective and pressure-proof caps shall be identified with the manufacturer's name or trademark, part number, and applicable military specification part number.

3.4.8 Contact arrangement and spacing. Contact arrangement and spacing shall be as specified (see 3.1).

3.4.9 Gaskets. O-ring gaskets shall be provided and shall conform dimensionally to MS28775 part numbers as specified (see 3.1).

3.4.10 Receptacle protective and pressure-proof cap assemblies. The penetrator receptacle protective and pressure-proof cap assemblies shall be as specified (see 3.1).

3.4.11 Hull penetrators and connectors. The penetrator and connector designs shall be as specified (see 3.1). The overall configuration of the hull penetrator body shall be such that it will allow installation in standard Navy submarine pressure-hull inserts as specified in MIL-C-24231 and MIL-C-24231/25.

3.4.12 Electrical wires and cables.

3.4.12.1 Conductor color code. Unless otherwise specified (see 3.1), the individual conductor color code numbers in cables shall be connected to matching number of socket contacts.

3.4.12.2 Cable type marking. The cable type shall be permanently marked on the molded plug or the cable within 6 inches of each plug. Cable manufacturer's repetitive cable identification markings are acceptable.

3.4.12.3 Receptacle wire marking. The ends of wires extending from the receptacle inserts shall be marked with the pin contact number to which they are attached.

3.4.12.4 Shielded cable insulation. Where shielded cables are molded in receptacle inserts, each shield shall be insulated from the other shields in the cable, except where the shields are connected to a common pin contact.

3.4.13 Soldering. The soldering and solder materials shall be in accordance with requirement 5 of MIL-STD-454.

3.4.14 Electrical receptacles. The electrical receptacles welded to the multiconnector hull penetrator body shall conform to the design requirements of MIL-C-24231.

3.4.15 Electrical plug assemblies. The electrical plug assemblies that mate with the electrical receptacles shall conform to the design requirements of MIL-C-24231.

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3.5 Interchangeability. The fiber optic and electrical hull penetrators, connectors, and accessories having the same specification part number shall be completely interchangeable with each other with respect to installation (physical) and performance (function) as specified herein.

3.6 Performance. Performance relates to the ability of the hull penetrator and its mated harness to provide satisifactory optical and electrical signal transmission(s) while providing a pressure-proof sea water barrier at the submarine pressure hull.

3.6.1 Insertion loss. Unless otherwise specified (see 3.1), when measured in accordance with 4.7.1, the maximum per channel insertion loss under all conditions shall be 5.0 decibels per series connector junction.

3.6.2 Discontinuity (optical). When measured in accordance with 4.7.2, there shall be no discontinuities during the test exposure. A discontinuity is considered here to be a reduction in signal strength of 2 decibels (37 percent) or more, for a duration of 20 nanoseconds or more unless otherwise specified (see 3.1).

3.6.3 Pulse distortion. When tested in accordance with 4.7.3, the rise and fall times of the transmitted optical pulse(s) shall be at least as long as, but no more than 0.5 nanosecond longer than the rise and fall times of the incident pulse(s). The duration of the transmitted optical pulse(s) shall be no greater than 0.5 nanosecond longer than the duration of the incident optical pulse(s).

3.6.4 Crosstalk. When tested in accordance with 4.7.4, the signal optical power level or sum of levels for devices with three or more channels of the passive channel or channels shall be below the output signal level of the active channel by at least 60 decibels.

3.6.5 Ambient optical pickup. When tested in accordance with 4.7.5, the optical power of the light from the optical port(s) (after accounting for cable and optical junction losses between the device and the detector) shall be less than -70 dBm.

3.6.6 Durability. When tested in accordance with 4.7.6, the mating plugs and receptacles shall show no evidence of excessive wear on engaging hardware, no uneven wear or galling on guide hardware, no metal chips or filings in the terminus or contact area(s), no damaged inserts, no other mechanical defects detrimental to connector operation. The optical performance requirements of table II shall be met.

3.6.7 Cable seal flexing. When tested in accordance with 4.7.7, pressure-proof molded cable seals shall prevent loss of environmental sealing or other damage which may impair the connector operation. The optical and electrical performance requirements of table II shall be met.

3.6.8 Hydrostatic pressure. When tested in accordance with 4.7.8, penetrators shall reveal no permanent dimensional changes, no cracking or crazing of insulation material, no permanent damage to cable sealing compounds, no leakage of hull penetrator to hull insert seals; no leakage through the penetrator internals; plug-to-receptacle seals; and cable-to-plug seals, displacement or rupture of connector shells, inserts, termini, contacts, fibers or sealing compounds, nor any other penetrator damage which may degrade the performance of the penetrator. The optical and electrical performance requirements of table II shall be met.

3.6.8.1 Hydrostatic pressure-secondary conductor and fiber seal(s) (when specified, see 3.1). When tested as specified in 4.7.8.1, there shall be no evidence of water leakage around or through the penetrator packing.

3.6.8.2 Hydrostatic pressure-primary termini or contact seals. When tested as specified in 4.7.8.2, there shall be no evidence of water leakage around or through the primary fiber optic cable seal.

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3.6.9 Thermal shock. When tested in accordance with 4.7.9, penetrators subjected to the high and low temperature extremes shall reveal no evidence of connector part dimensional change, no apparent loss of sealing capability, no surface or identification marking impairment, no coupling-thread binding or other evidence of mating or unmating incapability, and no other damage detrimental to the operation of the connector. The optical and electrical performance requirements of table II shall be met.

3.6.10 Vibration. When tested in accordance with 4.7.10, the hull penetrators and connectors shall exhibit no visual evidence of loosening of parts, relative motion between penetrator and cable parts or other damage which can produce physical distortion and result in fatigue of the mechanical parts. The optical and electrical performance requirements of table II shall be met.

3.6.11 High impact shock. When tested in accordance with 4.7.11, the penetrators and connectors shall not be damaged and there shall be no loosening of parts. The optical and electrical performance requirements of table II shall be met.

3.6.12 Hydrostatic pressure cycling. The penetrator shall be subjected to hydrostatic pressure cycling as specified in 4.7.12. There shall be no evidence of mechanical damage or water leakage. The optical and electrical performance requirements of table II shall be met.

3.6.13 Accelerated aging. When tested in accordance with 4.7.13, the hull penetrators subjected to these specified aging exposures shall exhibit no visual evidence of dimensional change, no opening or leaking of seals, no cracking or crazing of components or finishes, no identification marking impairment, nor any other effects detrimental to their operation. The optical and electrical performance requirements of table II shall be met.

3.6.14 Underwater explosion. The hull penetrators mated with outboard plugs shall not be damaged when subjected to the underwater explosion tests specified in 4.7.14. There shall be no evidence of water leakage into the hull penetrator or past the penetrator-to-hull insert seals as a result of this test. The penetrator assembly shall remain functional following the underwater explosion test, as specified (see 3.1). The optical and electrical performance requirements of table II shall be met.

3.6.15 Insulation resistance. When tested in accordance with 4.7.15, the insulation resistance shall be as follows:

a. Conductor to conductor: 2,000 megohms minimum.
b. Conductor to ground: 2,000 megohms minimum.
c. Shield to ground: 100 megohms minimum.

3.6.16 Electrical continuity. Connected contacts and conductors shall show no evidence of open circuits when tested in accordance with 4.7.16.

3.6.17 Dielectric withstanding voltage. When tested in accordance with 4.7.17, there shall be no evidence of breakdown or flashover when subjected to the test voltage of 1,600 volts root-mean-square, 60 hertz.

3.7 Workmanship. The hull penetrators, connectors, and accessories shall be dimensionally uniform and free of flaws that would degrade performance after installation, that would inhibit proper connection to interfacing elements and would otherwise yield an inferior product. The following shall be a minimum level of visual examination to be performed and is not intended to restrict other pertinent workmanship examinations deemed necessary by the contractor.

a. Penetrators, connectors, and accessories shall conform to the dimensional and interchangeability requirements of this specification.

b. Loose termini or contacts, poor molding, poor fabrication, loose materials, defective bonding, damaged or improperly assembled termini or contacts and physical defects in the hermetic seals shall not be permitted.

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c. Peeling or chipping of plating or finish, galling of mated parts, nicks or burrs of metal parts, and post molding warpage shall not be permitted.

d. Connector identification marking shall be legible and as specified (see 3.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Provision for test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection(s) shall be the responsibility of the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662 and as specified herein.

4.1.2 Inspection conditions. Unless otherwise specified (see 3.1), all inspections shall be made in accordance with the following conditions:

a. Temperature: +15°C to +35°C.
b. Relative humidity: 20 to 80 percent.
c. Barometric pressure: 550 to 800 millimeters of mercury.

4.2 Classification of inspections. The inspection and testing of penetrators covered by this specification are classified as follows:

a. First article inspection (see 4.3).
b. Materials inspection (see 4.4).
c. Quality conformance inspection (see 4.5).

4.3 First article inspection. First article inspection shall consist of the examinations and tests performed in the sequence specified in table I, on the test samples specified in 4.3.1. Control tests as noted in table II are to be conducted on the test samples before, during, and after the first article tests, as applicable.

4.3.1 First article samples. Two samples of each type for which first article inspection is desired shall be furnished for first article test (see 6.3). No failures shall be permitted to pass first article inspection.

4.3.1.1 Single connector hull penetrator - type I. Three single connector fiber optic hull penetrators shall be provided; two for sequence testing and one for underwater explosion testing by the Navy.

4.3.1.2 Multi-connector hull penetrator - type II. Two multi-connector fiber optic hull penetrators shall be provided for each size for which first article inspection is desired; one for sequence testing and one for underwater explosion testing by the Navy.

4.3.1.3 Connectors - types III and IV. Two complete fiber optic connectors of each shell size for which first article inspection is desired shall be provided for sequence testing.

4.3.1.4 Pressure-proof connector plug and receptacle caps - types V and VI. One pressure-proof connector plug and receptacle cap, as applicable, for each shell size desired for first article inspection shall be provided for sequence testing.

4.3.1.5 Multi-connector electrical and fiber optic hull penetrator - type VII. Two complete multi-connector electrical and fiber optic hull penetrators shall be provided for each size for which first article inspection is desired; one for sequence testing and one for underwater explosion testing by the Navy.

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TABLE I. First article inspection.

Examination or test	Requirement paragraph	Method paragraph	F.O. penetrator	F.O. penetrator 1/	F.O. connector	P-P cap plug-receptacle	Hybrid penetrator 2/	Hybrid penetrator 1/ 2/
Examination of product	3.1,3.3,3.4, 3.5 and 3.7	4.6	X	X	X	X	X	X
Insulation resistance	3.6.15	4.7.15					X	X
Continuity, electrical	3.6.16	4.7.16					X	X
Dielectric withstanding voltage	3.6.17	4.7.17					X	X
Insertion loss	3.6.1	4.7.1	X	X	X		X	X
Discontinuity (optical)	3.6.2	4.7.2	X	X	X		X	X
Pulse distortion	3.6.3	4.7.3	X	X	X		X	X
Crosstalk	3.6.4	4.7.4	X	X	X		X	X
Ambient optical pickup	3.6.5	4.7.5	X	X	X		X	X
Durability	3.6.6	4.7.6	X		X	X	X	
Cable seal flexing	3.6.7	4.7.7	X		X		X	
Hydrostatic pressure-static	3.6.8.1	4.7.8.1	X	X	X	X	X	
Thermal shock	3.6.9	4.7.9	X		X	X	X	
Vibration	3.6.10	4.7.10	X		X	X	X	
High impact shock	3.6.11	4.7.11	X		X	X	X	
Hydrostatic pressure-cycling	3.6.12	4.7.12	X		X	X		
Accelerated aging	3.6.13	4.7.13	X		X			
Underwater explosion	3.6.14	4.7.14		X				X
Post test examination	3.1,3.3,3.4, 3.5 and 3.7	4.6	X	X	X	X	X	X

1/ Underwater explosion test sample.

2/ Hybrid (electrical and F.O. contacts)

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TABLE II. Optical and electrical performance requirements.

Control test	Insertion loss	Discontinuity (optical)	Pulse distortion	Crosstalk	Ambient optical pickup	Product examination	Insulation resistance	Electrical continuity	Dielectric withstanding voltage
Durability	A					A		A	
Cable seal flexing	A					A		A	
Hydrostatic pressure-static	AD	D				A	BA	A	
Thermal shock	AD			A		A		A	
Vibration	A	D				A	A	AD	
High impact shock	A					A	A	AD	
Accelerated aging	A					A	A	A	
Hydrostatic pressure-cycling	D		A	A	A	A	BA	A	
Underwater explosion	A			A	A	A	BA	BA	A

A - Optical tests performed after the environmental test.

D - Optical tests performed during the environmental test.

B - Optical tests performed before the environmental test.

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4.3.1.6 Cable harness assemblies. The Government will supply cable harness assemblies necessary for conducting the hull penetrator, connector and pressure-proof plug cap tests. The harness assemblies will be fabricated in accordance with MIL-H-24626.

4.3.1.7 Disposition of sample units. The sample units which have been subjected to the first article inspection shall not be delivered on the contract.

4.4 Materials inspection. Materials inspection shall consist of certification, supported by verifying data, that materials specified (see 3.1 and 3.3) were used in fabricating the delivered penetrator, connectors.

4.4.1 Material control objective evidence. Each nickel-copper-aluminum component shipped shall be accompanied by material control objective evidence that the material conforms to all the requirements as specified (see 3.1). The following information shall be part of the objective evidence:

a. Assembly number.

b. Contract number.

c. Contractor's identity.

d. Serial number.

e. Material specification and heat treatments (with a statement to the effect that objective evidence is on file in support of material identification and heat treatments, traceable from the assembly serial number).

f. Direct reading of O-ring surface diameters with specified standard.

g. A notation that "32" rhr, material finishes are in accordance with the specification and are free of imperfections.

h. Dye penetrant test results of parts as applicable.

i. Signature or symbol of the contractor's inspector.

j. Signature or symbol of the U.S. Government's inspection representative.

4.4.1.1 Objective quality evidence. The contractor shall pack one copy of the objective quality evidence with the item and retain a duplicate copy on file. The cognizant inspection service is authorized to withhold shipment until the required objective quality evidence is in accordance with this specification. Failure to provide acceptable objective quality evidence shall be cause for rejection of the material.

4.4.2 Material control objective quality evidence in support of product quality. The material control objective quality evidence in support of product quality shall be maintained by the contractor for a minimum period of seven years.

4.4.3 Inspection of single component items. All items purchased separately, that is, in lieu of an entire assembly such as a penetrator or plug assembly, shall be controlled as specified in 4.4.1 through 4.4.2.

4.5 Quality conformance inspection. Quality conformance inspection shall be conducted on 100 percent of all production parts (see table III).

4.5.1 Inspection of products for delivery. Examinations and tests shall follow the test sequence of table III.

4.5.2 Failures. If any assembly fails to pass the quality conformance inspection, that unit shall be considered to have failed.

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4.5.3 Noncompliance. If any unit fails to pass the inspection, the contractor shall take corrective action on all materials or processes or both, as warranted, and on all units of product which are to be corrected and which were manufactured with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action acceptable to the acquiring activity has been taken. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.4 Packaging inspection. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-C-55330.

TABLE III. Quality conformance inspection.

Examination or test	Requirement	Inspection	F.O. penetrator	F.O. plug	F.O. receptacle	P-P cap	Hybrid penetrator
Examination of product	3.1, 3.3, 3.4, 3.5 and 3.7	4.6	X	X	X	X	X
Insulation resistance	3.6.15	4.7.15					X
Electrical continuity	3.6.16	4.7.16					X
Dielectric withstanding voltage	3.6.17	4.7.17					X
Insertion loss	3.6.1	4.7.1	X				X
Hydrostatic pressure-static	3.6.8.1	4.7.8.1	X		X	X	X
Post-test examination of product	3.1, 3.3, 3.4, 3.5 and 3.7	4.6	X		X	X	X

4.6 Examination of product. The as-received penetrators, connector assemblies, accessories, and piece parts shall be examined to ensure conformance with this specification as follows:

a. Applicable specification sheet (see 3.1).
b. Materials (see 3.3).
c. Design and construction (see 3.4).
d. Interchangeability (see 3.5).
e. Identification of product.
f. Finish.
g. Workmanship (see 3.7).

4.6.1 Hardness. The Brinell hardness of each nickel-copper-aluminum (K-Monel) component shall be checked to determine that the component is in the hardened condition. Hardness shall be checked at three locations, avoiding critical surfaces.

4.7 Methods of inspection. Cladding mode stripping devices shall be used when making optical measurements unless otherwise specified herein. The mode stripper shall be installed in the test circuit between the source and the connector. In devices with one, two, or three optical channels, optical measurements shall also be made with cladding mode-strippers inserted between the penetrator output (output connector) and the detector on each channel unless otherwise specified herein (see 3.1). In connectors with four or more channels, optical measurements shall be made simultaneously on three randomly selected channels unless otherwise specified herein.

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4.7.1 Insertion loss. The test for insertion loss is as follows. A suitable optical power source (an appropriate light emitting diode or injection laser diode) shall launch light into an optical circuit containing the device under test. The light may be static or modulated as appropriate. P_i and P_o are, respectively, the input and output optical signal powers. The insertion loss in decibels is 10 $\log_{10}$ (P_i/P_o). P_i and P_o are to be determined just to either side of the device under test so that, if applicable, the normal coupling losses associated with the device are included. Corrections shall be made for all losses along the optical circuit between the points where the optical powers are actually measured, and the points where P_i and P_o are to be determined. Since the source, detector, optical waveguide and interconnection characteristics of the optical circuit may change during the course of the test, some control or correction is required to ensure that the optical loss calculated is due solely to the insertion of the device under test. This may be accomplished by providing optical circuit stability or by using data from a reference optical transmission path to correct the data obtained from the transmission path containing the device under test. In devices with one, two or three optical channels the insertion loss shall be measured on each channel. In connectors with four or more channels the insertion loss shall initially be made on all channels. On subsequent tests the insertion loss shall be measured on three channels, or twenty percent of all channels, whichever has the larger number of channels. The measurements shall be made on randomly selected channels that have not previously been tested since the initial test, if possible (see 3.6.1). Insertion loss should be tested for all channels during the Quality Conformance Inspection and at the conclusion of the First Article Tests.

4.7.2 Discontinuity (optical). The test for discontinuity shall be as follows. One optical channel of the device under test shall be connected through short optical test cables to an appropriate optical signal source and detector. Unused ports shall be capped. The source shall produce a constant level, static signal easily detected by the detector. The output of the detector shall be monitored for discontinuities while the device under test is subjected to a physical test. The detector and monitoring equipment shall possess sufficient sensitivity and high-frequency response to detect discontinuities in the optical signal. The monitoring equipment shall also include suitable recording capability such as provided by an oscilloscope and camera so that a permanent record of the test data can be obtained (see 3.6.3).

4.7.3 Pulse distortion. The test for pulse distortion shall be as follows. The optical ports of the device under test shall be connected through short optical test cables to an optical signal source and detector. Unused ports shall be capped. The signal source shall provide a series of optical pulses having maximum rise times and fall times of 1 nanosecond, and pulse durations of approximately 5 nanoseconds. The rise and fall times shall be measured between the 10 and 90 percent optical power levels; and the duration between the 50 percent levels. The transmitted optical pulses shall be compared with the incident optical pulses (or other reference pulses known to be of the same shape as the incident pulses). The detector and monitoring equipment shall possess sufficient sensitivity and frequency reponse to suitably monitor both the incident (or reference) pulses and the transmitted pulses.

4.7.4 Crosstalk. The test for crosstalk is as follows: The ports of the circuits under test shall be connected to short test cables or opaquely capped as appropriate. The input port to one optical channel (active channel) is connected via its test cable to an optical signal source, the signal of which may be either continuous or amplitude modulated as appropriate. The output port(s) of the other channels (passive channel(s)) are connected to test cable(s). The other input ports shall be opaquely capped. The output signal power from both the active and passive channels shall be measured. When there is more than one passive channel, the output of all passive channels shall be summed. This test shall be repeated, making such remaining input port the active channel (see 3.6.4).

4.7.5 Ambient optical pickup. The test for ambient optical pickup shall be as follows. The optical ports of the device under test shall be either opaquely capped or connected to short optical test cables as appropriate. The far ends of these test cables shall be either capped or connected to suitable optical power monitoring instruments. Mode stripping shall not be employed. The device under test is (as far as practical) isotropically and homogeneously bathed in simulated sunlight as defined in method 505 of MIL-STD-810. The light shall be broad spectrum with infrared,

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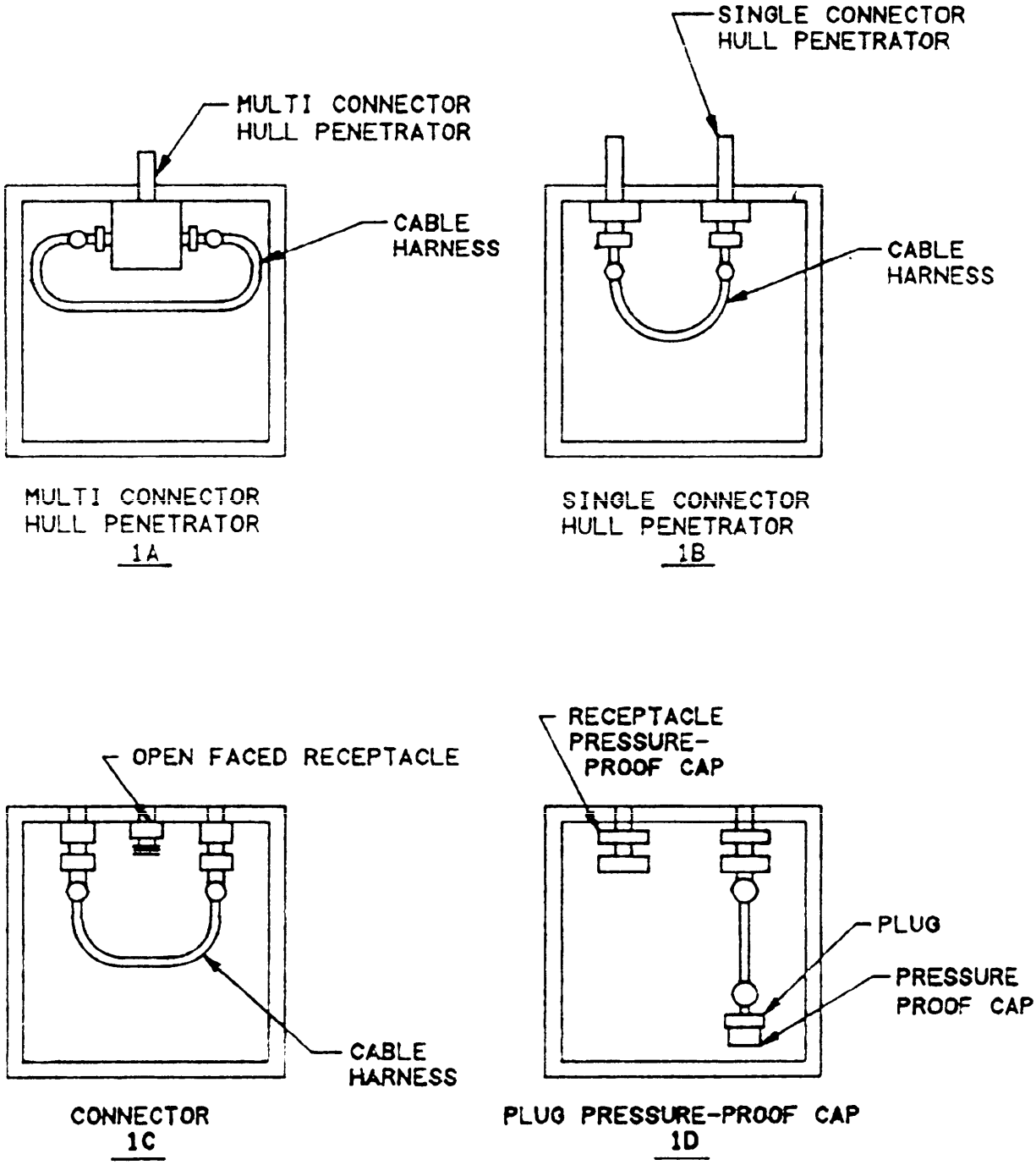


FIGURE 1. Hydrostatic pressure test setup.

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visible, and ultraviolet components and shall illuminate the device under test with an irradiance (power density) of 112 ±5 milliwatts per square centimeter. Light emanating from the optical port(s) of the device under test shall be measured by optical power monitoring equipment having a broad spectral response compatible with the source (see 3.6.5).

4.7.6 Durability. The connector assemblies or accessories shall be completely mated and unmated 100 times at a maximum rate of 50 cycles per hour with the plug coupling ring operated in a manner to simulate actual service (see 3.6.6).

4.7.7 Cable seal flexing. The connector assembly shall be tested in accordance with method 2017 of MIL-STD-1344, procedure I (see 3.6.7).

4.7.8 Hydrostatic pressure.

4.7.8.1 Hydrostatic pressure static. The hull penetrators shall be installed in a pressure vessel with the external section of the penetrator facing into the chamber as shown on figure 1. The pressure vessel test medium shall be tap water. The hull penetrator shall then be subjected to each step of the hydrostatic pressure test schedule specified on table IV. The optical and electrical performance tests as specified in table II shall be conducted before and following the pressure tests. (See footnote in table IV for test schedule for quality conformance tests). (See 3.6.8.1). For first article testing the penetrator shall be tested in the mated and unmated condition.

4.7.8.2 Hydrostatic pressure-secondary conductor and fiber seal(s). The secondary conductor seal shall be subjected to each step of the hydrostatic pressure test as specified in table IV. The secondary seal packing shall be installed in the pressure vessel with its outboard side facing to the chamber. The pressure vessel test medium shall be tap water. The secondary seal packing shall be mounted in a test flange having identical internal dimensions as those specified in the penetrator body design (see 3.6.8.2).

4.7.9 Thermal shock. The penetrator assembly, including mated plug assembly, shall be subjected to the thermal shock test as specified in method 1003 of MIL-STD-1344, test condition A, with the temperature condition of table V. The optical and electrical performance tests as noted in table II shall be conducted following the thermal shock test (see 3.6.9).

TABLE IV. Hydrostatic pressure sequence.

Step	Pressure lb/in^2(Pa)			Hold time (at high pressure)
	Low	High	Low	
1	0	20 (.140)	0	5 minutes
2	0	20 (.140)	0	5 minutes
3	0	20 (.140)	0	5 minutes
4	0	100 (.70)	0	5 minutes
5	0	250 (1.75)	0	5 minutes
6	0	500 (3.5)	0	5 minutes
7	0	1000 (7.0)	0	24 hours 1/

1/ 2 hours for quality conformance test.

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TABLE V. Thermal shock test conditions.

Step	Temperature - °C
1	$-30\ ^{+0}_{-5}$
2	$+4 \pm 5$
3	$+60\ ^{+5}_{-0}$
4	$+15 \pm 5$

4.7.10 Vibration. The hull penetrator assembly, shall be vibrated in accordance with MIL-STD-167-1; type I classification. The mating plug coupling nut(s), shall be held in place by the normal locking device. Cables shall be supported on a stationary frame not closer than 12 inches (304.8 mm) from the connectors. The optical performance tests as noted in table II shall be conducted before and following the vibration test. Optical discontinuity tests shall be conducted during the vibration tests. There shall be no interruption of electrical continuity longer than 1 microsecond during the test. Insulation resistance and continuity tests shall be conducted following the test (see 3.6.10). The electrical tests are required only as specified (see 3.1).

4.7.11 High impact shock. The hull penetrator assembly shall be tested in accordance with MIL-S-901, grade A, class I, type B. The plug coupling rings shall be held in place by normal locking means. The cable(s) shall be unsupported for a minimum distance of 12 inches (304.8 mm) from plug or penetrator. The optical performance tests as noted in table II shall be conducted before and following the high impact shock test. Optical discontinuity tests shall be conducted during the shock test. There shall be no interruption of electrical continuity longer than one microsecond during the test. Insulation resistance and continuity tests shall be conducted following the test (see 3.6.11).

4.7.12 Hydrostatic pressure-cycling. The hull penetrator assembly outboard cables shall be connected. The penetrator shall be installed in a pressure vessel with the external section of the penetrator facing to the chamber as shown on figure 1. The pressure vessel test medium shall be tap water. The penetrator assembly shall be subjected to 2,000 cycles of 1,000 lb/in^2 hydrostatic pressure. The test cycle shall be as follows: five minutes at 0 lb/in^2, 10 minutes to rise to the test pressure, 5 minutes hold at test pressure and 10 minutes to drop to 0 pressure. The time duration of each pressure cycle shall be 30 minutes plus 1 minute minus 0 minute. Tap water temperature for the first 500 cycles shall be 0°C-2°C; temperature for the next 1,000 cycles shall be 4.5°C-30°C; the last 500 cycles shall be conducted at a 30°C +3°C-0°C tap water temperature. The optical performance tests as specified in table II shall be conducted before and following the hydrostatic pressure tests. Insertion loss shall be measured during the test (see 3.6.12).

4.7.13 Accelerated aging. The hull penetrator assembly shall be subjected to an accelerated aging test for a period of 1,000 hours. The temperature shall be 60°C +5°C, -0°C with a relative humidity of 95 percent. The outboard cable harnesses shall be installed in the hull penetrator(s). The inboard fiber optic cables shall exit the test chamber and be sealed to the chamber wall at the exit point. The optical and electrical performance tests as noted in table II shall be conducted following the aging tests (see 3.6.13).

4.7.14 Underwater explosion. The hull penetrator and mated outboard plug assemblies shall be subjected to depth charge tests. These tests shall be performed by the Navy in accordance with their standard test methods which are classified (see 6.2). The optical and electrical performance tests as specified in table II shall be conducted before and after the underwater explosion test (see 3.6.14).

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4.7.15 Insulation resistance. Insulation resistance shall be tested in accordance with method 3003 of MIL-STD-1344 to determine conformance with 3.6.15. The resistance shall be measured between all adjacent contacts, conductors, and shields and between the penetrator, and each contact, conductor and shield.

4.7.16 Electrical continuity. All assembled electrical penetrators shall be checked for continuity with a standard circuit tester (see 3.6.16).

4.7.17 Dielectric withstanding voltage. Dielectric withstanding voltage shall be tested in accordance with method 3001 of MIL-STD-1344. The test voltage shall be applied between all adjacent contacts or conductors and between all contacts or conductors and the metal housing (see 3.6.17).

4.8 Cleaning samples.

4.8.1 Normal cleaning. Normal cleaning shall consist of removing salt contaminants first with warm distilled water. Excess water shall be blown dry with oil-free dry air. The air shall be warmed to 52°C. In addition, the component shall be washed with technical grade methyl alcohol and parts scrubbed with a small bristle brush. Excess alcohol can be blown dry with dry, oil-free air. Normal cleaning may be used prior to control tests to remove surfaces contaminants from insulator and contact and terminus surfaces.

4.8.2 Forced cleaning. Air drying at temperatures elevated above 52°C and requiring more than three of these cycles specified in 4.8.1 is considered force and not permitted.

4.8.3 Contact cleaning. The contact cleaning procedure shall be approved by NAVSEA.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330.

6. NOTES

6.1 Intended use. The fiber optic and electrical hull penetrators, pressure-proof connectors, and accessories described in this specification are intended for use on submarines.

6.2 Ordering data. Acquisition documents shall specify the following:

a. Title, number and date of this specification.

b. Title, number and date of the applicable specification sheet (see 3.1) and the complete part number.

c. Special wiring requirements.

d. Length of penetrator in board cable required if other than as specified or listed in the applicable specification sheet (see 3.1).

e. Explosion test data to be supplied to contractor by the Government (see 4.7.14).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definitions

6.4.1 Body. The cylindrical pressure-proof housing on which the connectors are mounted on the outboard side. The body is usually O-ring sealed to a hull insert and retained with a nut located on the inboard side of the hull.

6.4.2 Fiber optic cable harness assembly. A fiber optic cable with connector plugs or receptacles connected and sealed to each end of the cable.

6.4.3 Connector plug. The plug is usually that portion of the connector set which is affixed to the cable.

6.4.4 Coupling ring. A device fitted to the plug assembly which engages and disengages the plug to the receptacle.

6.4.5 Designation plates. A hull penetrator identification plate which denotes receptacle position numbers, penetrator hole number and specification number.

6.4.6 End flanged bolts receptacle. A receptacle having a flange at the rear of the receptacle body to provide for O-ring sealing and bolting the receptacle to the penetrator or component housing.

6.4.7 Flanged welded receptacle. A receptacle having a flange at the mid-section of the receptacle body to provide for welding the flange to the penetrator or component housing; thus, shortening of the exposed portion of the receptacle from the housing wall.

6.4.8 Fiber optic hull penetrator. A pressure-proof device which is designed to allow the passage of fiber optic signals through the pressure hull of a submarine or vehicle. The penetrator seals the fiber optic termini and cables against sea water entry at the pressure hull.

6.4.9 Gland nut. A threaded member which, when tightened, compresses a rubber packing to provide a fiber optic cable seal.

6.4.10 Hull insert. A structural member used to reinforce the hull plate in the immediate vicinity of a hole through the hull. It is most often a thick walled cylinder welded into the hold of the submarine pressure hull.

6.4.11 In-line receptacle. A receptacle wires and molded to a cable.

6.4.12 Locknut receptacle. A receptacle having an O-ring sealing flanged at the front of the receptacle and threads at the rear end to allow fastening the receptacle to the housing with a locknut.

6.4.13 Mid-flange bolted receptacle. A receptacle having a flange at the mid-section of the receptacle body to provide for O-ring sealing and volting the receptacle to the penetrator or component housing.

6.4.14 Molded boot. The molded or vulcanized member of a harness forming the seal between the fiber optic cable and the fiber optic connector.

6.4.15 Molded insert assembly. A cylindrical epoxy housing used in the receptacles to seal, locate, and insulate the contacts in termini to the receptacle body. The insert assembly has a groove to house the O-ring gasket which seals the insert to the receptacle body.

6.4.16 Multi-connector fiber optic hull penetrator. A hull penetrator design which houses two or more pressure-proof fiber optic connector receptacles in a turret fashion around the outboard side of the penetrator body.

6.4.17 Nut. A large metal nut located on the inboard side of the penetrator body to retain the penetrator in the pressure hull.

6.4.18 Packing. A cylindrical elastomeric seal containing holes through which the fiber optic cables pass through to provide a secondary cable seal.

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6.4.19 Pressure-proof receptacle cap. A cover fitted to a receptacle whose primary function is to provide protection against full submergence sea water pressure.

6.4.20 Pressure-proof plug caps. A cover fitted to a plug whose primary function is to provide protection against full submergence sea water pressure.

6.4.21 Pressure plug cap. A nonpressure-proof cover fitted to a plug whose primary function is to protect against mechanical damage and contamination by dirt or foreign objects.

6.4.22 Protective receptacle cap. A nonpressure-proof cover fitted to a receptacle whose primary function is to protect against mechanical damage and contamination by dirt or foreign objects.

6.4.23 Receptacle. The receptacle is normally the fixed member of the connector set.

6.4.24 Receptacle style. The general configuration of a receptacle which better suits it to one type of mounting to a housing than another. The various styles are: welded; flange welded; in-line; mid-flanged bolts; end flanged bolted; locknut and union.

6.4.25 Retainer plates. Discs situated on both inboard and outboard faces of a packing which tends to distribute the thrust of the gland nut evenly over the packing to provide fiber optic cable sealing. The plates present packing extrusion into cavities which would allow seal loss.

6.4.26 Right angle plug. A plug connector having the cable wires and molded at a 90° angle to the axis of the plug shell.

6.4.27 Single connector fiber optic hull penetrator. A hull penetrator design which houses only one pressure-proof fiber optic connector receptacle on the outboard side of the penetrator body.

6.4.28 Straight plug. A plug connector having the cable wired and molded on the same axis of the plug shell.

6.4.29 Union receptacle. A receptacle having female cavities on both ends of the receptacle to provide for mating plug connectors to each end of the receptacle.

6.4.30 Washer. A cylindrical device located between the nut and the hull insert to provide a sealing gasket surface for the secondary O-ring gasket which seals the penetrator to the hull insert.

6.4.31 Welded receptacle. A receptacle which is affixed to a housing by welding at the rear of the receptacle.

6.4.32 Multi-connector electrical fiber optic hull penetrator. A pressure-proof device which is designed to allow the passage of electrical and fiber optic signals through the pressure hull of a submarine or vehicle. The penetrator seals the fiber optic termini and electrical contacts and cables at the pressure hull. The penetrator houses one or electrical and fiber optic pressure-proof connector receptacles in a turret fashion around the outboard side of the penetrator body.

6.4.33 Insertion loss test. As used herein, insertion loss is the total optical signal power loss in an optical circuit caused by insertion of the optical device in question into the optical circuit. The purpose of this test is to confirm the device does not excessively attenuate the optical signal.

6.4.34 Discontinuity test. As used herein, signal discontinuity refers to measuring the presence of relatively sharp, short duration interruptions or "drop-outs" of the optical signal(s) transmitted by the optical device in question. The purpose of this test is to confirm the device's signal discontinuities are not excessive when subjected to physical stresses.

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6.4.35 Analog modulation test. As used herein, analog modulation refers to dynamic amplitude modulation of the optical signal(s) transmitted by the optical device in question. The purpose of this test is to confirm the device does not excessively modulate the optical signal when subjected to physical stress tests.

6.4.36 Pulse distortion test. As used herein, pulse distortion refers to the change in shape or short duration optical pulses transmitted by the fiber optic device in question. The purpose of this test is to verify the device does not excessively distort the shape or change the duration of the transmitted optical pulses.

6.4.37 Crosstalk test. Crosstalk as used herein refers to unwanted, coupled optical energy from an optical circuit (the active channel) into another optical signal circuit or group of signal circuits (the passive channel(s)). The purpose of this test is to ensure that the circuits of the device in question have adequate optical channel isolation.

6.4.38 Ambient optical pickup test. As used herein, ambient optical pickup refers to optical interference of the optical data signal(s) in the device in question by ambient optical power incident upon its exterior. The purpose of this test is to confirm the ability of the device to exclude ambient optical power incident upon its exterior from its optical signal circuit(s).

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Custodians:
Navy - SH

Review activities:
Navy - AS, EC
DLA - ES

User activity:
Navy - CG, MC

Agent:
DLA - ES

Preparing activity:
Navy - SH

(Project 6060-N011)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-24628(NAVY)	FIBER OPTIC, SUBMARINE GENERAL SPECIFICATION FOR PENETRATORS, HULL, CONNECTORIZED, CONNECTORS, PRESSURE-PROOF,

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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